Sleep Patterns and Mental Wellbeing in Māori and Pasifika Youth: An Analysis Using the Growing Up in New Zealand Study Data

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Date of Submission: 30 October 2025

Abstract
This study analyses sleep patterns and their relationship with mental wellbeing among Māori and Pasifika youth using longitudinal data from the Growing Up in New Zealand (GUiNZ) study. Data from ages 2 to 12 were examined to explore sleep duration, timing, and difficulties alongside mental health outcomes—depression, anxiety, and resilience—at age 12. Results show a clear developmental decline in sleep, with Māori and Pasifika children consistently sleeping less than Sole Europeans, largely due to socioeconomic and household factors such as crowding, food insecurity, and family structure. Shorter sleep duration was associated with higher depression and anxiety scores, though early childhood sleep did not predict later depression, suggesting that psychosocial influences in later childhood play a stronger role. Grounded in Kaupapa Māori and Pacific frameworks, the study emphasises culturally responsive and equitable approaches to improving sleep health through community-based interventions and integrated mental health policies.

Keywords: *Sleep patterns, mental wellbeing, Māori youth, Pasifika youth, longitudinal analysis, GUiNZ study*

Acknowledgements

I extend my deepest gratitude to my supervisors, Dr Matthew Pawley, Dr Rebecca Evans, and Dr Nicola Ludin, for their expert guidance, insightful feedback, and continual encouragement throughout the course of this project. Their mentorship and dedication have been instrumental in shaping both my analytical approach and academic growth.

I am sincerely thankful to the Growing Up in New Zealand (GUiNZ) study team at the University of Auckland for providing access to the dataset and ensuring the ethical integrity of this research. Special appreciation goes to Terry Li for his technical assistance in accessing and managing the GUiNZ data through AWS.

I would also like to acknowledge Māori and Pasifika researchers, including Dr Roannie Ng Shiu and Associate Professor Aylana Mikaere-Hall, whose cultural expertise and guidance ensured that this study remained grounded in the principles of Kaupapa Māori and Pacific research methodologies.

Finally, I extend heartfelt thanks to the GUiNZ families whose participation makes this research possible. Their generosity and trust continue to drive efforts to enhance the wellbeing of children and whānau across Aotearoa New Zealand.

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Introduction

Sleep

Sleep is a fundamental determinant of physical, emotional, and cognitive development in childhood and adolescence. Adequate and consistent sleep supports learning, emotional regulation, and resilience, while chronic sleep deprivation is associated with behavioural difficulties, anxiety, and depressive symptoms (Galland et al., 2020). In Aotearoa New Zealand, recent studies indicate that a substantial proportion of children and adolescents are not meeting recommended sleep durations, with marked disparities across ethnic and socioeconomic groups(Fleming et al., 2020; Paine et al., 2022). Recent national evidence from the Growing Up in New Zealand (GUiNZ) study reinforces these disparities: children's 24-hour activity and sleep patterns differ significantly by gender, ethnicity, and household deprivation, with only 62.5 percent meeting the sleep guideline and just 10.6 percent meeting the combined 24-hour Movement Guidelines (Hedayatrad et al., 2022).

Māori and Pasifika youth, in particular, experience disproportionately high rates of poor sleep and mental health challenges compared with Sole European peers (Fleming et al., 2022). These inequities are not simply biological or cultural but reflect broader structural factors such as housing quality, crowding, and social deprivation (Morton et al., 2013). Cultural practices such as co-sleeping and collective family living—common within Māori and Pasifika whānau—also influence sleep patterns in ways that differ from Western norms (Carter et al., 2025). Despite this, most existing sleep research has been Western-centric, often overlooking the cultural and contextual dimensions that shape sleep experiences and mental wellbeing (Paine et al., 2022).

The Growing Up in New Zealand (GUiNZ) longitudinal study, led by (Morton et al., 2013), provides a unique opportunity to explore these dynamics within a large, ethnically diverse cohort of children born between 2009 and 2010. By analysing repeated measures of sleep behaviour from early childhood through pre-adolescence, alongside validated assessments of mental wellbeing at age 12, this project investigates how early sleep trajectories differ by ethnicity and household context, and how they relate to later emotional outcomes.

Grounded in Kaupapa Māori principles and Pacific research methodologies, this research adopts a strengths-based and culturally responsive approach. It seeks not only to identify patterns of risk but also to highlight the protective roles of whānau relationships, cultural connectedness, and collective wellbeing (Carter et al., 2025). Through this lens, the project aims to contribute evidence that informs equitable and culturally grounded health interventions—supporting healthier sleep and improved mental wellbeing among Māori and Pasifika youth in Aotearoa New Zealand.

Overview of the Organisation

The Growing Up in New Zealand (GUiNZ) study, led by the University of Auckland, is Aotearoa New Zealand's largest and most comprehensive longitudinal investigation of child development. Established in 2009, it follows a birth cohort of 6,846 children born between 2009 and 2010, designed to be nationally representative and ethnically diverse (Morton et al., 2013). The cohort includes approximately 1,300 Māori and 950 Pasifika participants, providing a robust basis for examining cultural and socioeconomic influences on wellbeing. Data collection spans from the antenatal period to age 12 years, encompassing domains such as health, sleep, education, family dynamics, and cultural identity across multiple waves.

The GUiNZ study aims to generate evidence that informs equitable and effective social and health policies, with a particular focus on reducing disparities among children from historically marginalised communities. This project contributes to that mission by analysing longitudinal sleep and wellbeing patterns within the GUiNZ cohort. Conducted in partnership with the GUiNZ research team, the analysis aligns with Te Tiriti o Waitangi principles and GUiNZ's Kaitiaki Framework, ensuring cultural safety and respect for Māori and Pasifika worldviews. Collaboration with indigenous researchers strengthens the integrity, interpretation, and policy relevance of the findings.

Project Background and Objectives

Sleep and mental wellbeing are deeply interconnected aspects of child development, particularly within Māori and Pasifika communities, who experience disproportionately higher rates of anxiety, depression, and sleep disturbances (Fleming et al., 2020; Paine et al., 2022).The first 1000 days of life—from conception to age two—are critical for shaping lifelong health, cognitive and social development, highlighting the importance of early environments, family stability, and equitable access to support (Morton et al., 2024). National evidence from the Youth19 Rangatahi Smart Survey indicates that 39% of New Zealand youth sleep less than recommended, while 57% report poor sleep quality, which is strongly associated with mental health difficulties (Galland et al., 2020). Despite this, limited longitudinal research explores how cultural practices—such as co-sleeping and extended family environments—shape sleep patterns and wellbeing outcomes across childhood.

This project investigated the longitudinal relationship between sleep and mental wellbeing among Māori, Pasifika, and Sole European children using data from the

Growing Up in New Zealand (GUiNZ) study.
The research was structured around three key questions:

- RQ1: Examine how sleep patterns (duration and reported difficulties) from ages 2 to 12 differ across ethnic groups, identifying developmental trajectories and cultural influences on sleep behaviour.
- RQ2: Determine whether early childhood sleep patterns (ages 2–8) predict mental wellbeing outcomes at age 12—including depression, anxiety, and resilience—highlighting potential early indicators of later psychological health.
- RQ3: Assess how socioeconomic and contextual factors (e.g., deprivation, household crowding, urban–rural residence) influence the relationship between sleep and mental wellbeing, revealing structural inequities and modifiable determinants.

The overarching objective is to provide evidence-based, culturally responsive recommendations to improve sleep and mental health equity for Māori and Pasifika youth. Secondary objectives include identifying mediating effects of family structure, socioeconomic status, and cultural practices to inform policy design and targeted interventions that promote holistic wellbeing.

Literature Review

Background and Size of Effects

Mental health disorders such as anxiety and depression (AD) are among the leading causes of disability in youth worldwide, accounting for up to 70% of disability-adjusted life years (Patel et al., 2007). In Aotearoa New Zealand, national surveys report that 39% of young people sleep less than the recommended amount for their age, while 57% experience poor sleep quality(Fleming et al., 2022; Galland et al., 2020). Sleep disturbance is an established early risk factor for emotional and behavioural problems, particularly anxiety and depression, during adolescence (Carpenter et al., 2021).

Ethnic disparities are pronounced: Māori and Pasifika youth show higher rates of anxiety, depression, and poor sleep compared with Sole European peers (Clark et al., 2014; Paine et al., 2022; Ruhe et al., 2022). For instance, Pasifika youth have a 1.4-fold higher likelihood of experiencing anxiety or depression, with nearly one-quarter affected by depressive symptoms (Fleming et al., 2022; Oakley Browne et al., 2010). Moreover, specific populations such as transgender students and those in alternative education settings face even greater vulnerability, including elevated rates of depressive symptoms, self-harm, and suicide attempts (Clark et al., 2014; Denny et al., 2004).

Despite this burden, most evidence-based treatments such as antidepressants and psychotherapy demonstrate modest or inconsistent effects (Cipriani et al., 2016; Xiang et al., 2022). Increasing attention is now given to upstream determinants—such as sleep and circadian rhythm disruption (Slopen et al., 2016) broader environmental conditions that shape child wellbeing (Howden-Chapman et al., 2021).

The New Zealand Context

Aotearoa New Zealand's diverse population continues to experience inequities rooted in colonisation, social deprivation, and structural disadvantages. Māori and Pasifika youth remain disproportionately affected by poor sleep and mental health outcomes, not because of ethnicity itself, but due to systemic inequities and unequal access to resources (Howden-Chapman et al., 2021; Muller et al., 2024; Paine et al., 2022). Overcrowded or substandard housing—conditions more prevalent among Māori and Pasifika whānau—contribute to disrupted sleep and increased psychological distress (Howden-Chapman et al., 2021).

Longitudinal and national surveys such as Youth'12, Youth19, and the Adolescent Health Survey show a consistent increase in depressive symptoms among rangatahi Māori and Pasifika between 2012 and 2019, alongside widening wellbeing disparities (Clark et al., 2014; Fleming et al., 2022). (Clark et al., 2022) emphasised that advancing equity for Māori youth is central to Aotearoa's social wellbeing. Concurrently, research has underscored the strengths within Māori and Pasifika communities—particularly whānau support, collective identity, and cultural connectedness—as protective factors against adversity and distress (Carter et al., 2025; Williams et al., 2018).

Kaupapa Māori and Pacific research frameworks challenge deficit-based narratives by emphasising cultural safety, relationality, and collective wellbeing (Suaalii-Sauni & Fulu-Aiolupotea, 2014). These paradigms align with the GUiNZ Kaitiaki Principle, ensuring that Indigenous and Pacific worldviews guide both research interpretation and policy implications.

Cultural and Environmental Influences

Cultural and environmental factors substantially shape sleep behaviours and interpretations of wellbeing. Within Māori and Pasifika whānau, practices such as co-sleeping, room-sharing, and intergenerational caregiving are common and often grounded in values of whanaungatanga, security, and collective care (Carter et al., 2025; Morton et al., 2015). While Western literature has historically pathologised these

practices, emerging evidence suggests they may enhance emotional bonding and resilience when situated within stable family environments (Paine et al., 2022).

The Growing Up in New Zealand (GUiNZ) study offers an ideal framework for exploring these dynamics. Designed to reflect Aotearoa's ethnic diversity, GUiNZ follows more than 6,800 children from birth to adolescence (Morton et al., 2013; Morton et al., 2015). Findings from GUiNZ and related projects (e.g., Moe Kura) show that sleep practices, sleep duration, bedtime, and wake-time patterns vary by ethnicity and are influenced by both material conditions and cultural norms. Notably, disparities widen with age: Māori and Pasifika adolescents experience shorter sleep and later bedtimes than Sole Europeans, even after adjusting for socioeconomic status (Muller et al., 2024).

Summary and Research Gap

The reviewed literature consistently demonstrates that sleep plays a fundamental role in children's emotional, cognitive, and psychological development (Carpenter et al., 2021; Galland et al., 2020). Within Aotearoa, Māori and Pasifika youth continue to experience inequitable sleep and mental health outcomes, shaped by socioeconomic deprivation, housing conditions, and limited access to culturally safe health services (Fleming et al., 2022; Howden-Chapman et al., 2021).

However, most prior studies have been cross-sectional and grounded in Western biomedical paradigms that overlook Indigenous and Pacific worldviews. Few have examined how early-life sleep patterns predict later wellbeing using a culturally grounded, longitudinal framework. This gap underscores the need for research that integrates structural, social, and cultural determinants across developmental stages.

Accordingly, the present study draws on longitudinal GUiNZ data to examine how early childhood sleep trajectories (ages 2–8) relate to mental wellbeing at age 12 among Māori and Pasifika children. By combining Kaupapa Māori and Pacific research methodologies with advanced statistical and machine-learning approaches, the study aims to generate evidence that supports equitable, whānau-centred sleep-health interventions for Aotearoa's diverse communities.

Methodology

Data Source and Variables

Data were drawn from GUiNZ waves at ages 2, 4, 8, and 12 years. Sleep variables included night-time and daytime duration, total sleep, bedtime/wake time, location (e.g., co-sleeping), and parental-reported difficulties. Mental wellbeing at age 12 was assessed using CES-D-10 (depression), PROMIS T-Score (anxiety), CD-RISC (resilience), and Strengths and Difficulties Questionnaire (SDQ) for internalizing

symptoms. Covariates included ethnicity (Māori, Pasifika, Sole European), socioeconomic deprivation (NZDEP), household crowding, food insecurity, family structure, and rurality. The sample focused on Māori (n≈1,300) and Pasifika (n≈950), with Sole European as the reference group.

Variables and Measures

Table 1 summarises the primary variables analysed in this study, including sleep, psychological wellbeing, and sociodemographic indicators. Each variable was drawn from the Growing Up in New Zealand (GUiNZ) longitudinal dataset and measured at one or more developmental waves (ages 2, 4, 8, and 12 years). Sleep variables capture both quantity and quality dimensions, while psychosocial indicators assess emotional wellbeing and family context. All variables were cleaned, harmonised, and coded according to GUiNZ metadata specifications before analysis.

Category	Variable Name (Dataset Code)	Description / Measure	Scale or Coding	Age Wave(s)
Sleep Behaviour	SLP1_Hours_Y2CM, SLP1_hr_m45CM, SLP1_HR_Y8CM, SLP1_HR_Y12CM	Night-time sleep duration (hours per night).	Continuous (hours)	2, 4, 8, 12 years
	SLP2_Hours_Y2CM, SLP2n_hr_m45Cm	Daytime nap duration.	Continuous (hours)	2, 4 years
	SLP_TOTAL_HOURS (derived)	Combined day + night sleep duration.	Continuous (hours)	2–12 years
	SLP10_Y12CM	Mother-reported sleep quality.	1 = No problem, 2 = Small, 3 = Moderate, 4 = Large	12 years
	SLP16_Y12C	Child-reported sleep quality.	1= Very well,2= Fairly well, 3= fairly badly, 4= Very badly	12 years
Psychological Wellbeing (Age 12)	DEPRESS_SCORE_10_Y12C	CES-D-10 Depression Score (0–30).	Continuous	12 years
	PAS_T_SCORE2_Y12C	PROMIS Anxiety T-Score.	Continuous	12 years
	CDRISC_SUM_Y12C	Connor-Davidson Resilience Scale (CD-RISC-10).	Continuous (0–40)	12 years
Household / Socioeconomic Context	OECD_HH_INCOME_Y12M	Household income (OECD-equivalised).	Continuous (NZD)	12 years
	HHST_Y12M	Household structure (nuclear, extended, non-kin).	Categorical	12 years
	AGG_FIS_CAT_Y12M	Food Insecurity Category.	1 = Severe, 2 =	12 years

			Moderate, 3 = Secure	
	HHTENURE_Y12M	Housing tenure.	Own, Private rental, public housing, Other	12 years
	CROWDINGGP_Y12M	Household crowding index (people per bedroom).	Low (≤ 1), Medium (1–2), High (> 2)	12 years
	RURALITY_BIN_UR2018_Y12C	Urban–Rural classification (2018 UR profile).	Urban = 0, Rural = 1	12 years
	NZDEP2018_5_Y12C	New Zealand Deprivation Index (Deciles 1–5).	1 = least deprived	12 years
Demographic Variables	EXT_PROETH_y12C	Primary child ethnicity (self- or parent-identified).	Māori / Pasifika / Sole European (reference)	12 years
	TRANS_NB_CAT_Y12C	Child gender identity.	Boy, Girl, Trans/NB (aggregated for analysis)	12 years
	ChildID	Unique participant identifier for longitudinal linking.	Nominal (unique ID)	All waves

Table 1 Summary of Variables and Measures Used in the study

Analytical Approaches

Descriptive and inferential analyses were conducted to examine children's sleep patterns, identify demographic and environmental predictors, and evaluate longitudinal associations with mental wellbeing outcomes. The analysis combined traditional inferential statistics and contemporary machine-learning methods to provide both explanatory and predictive insights. Linear and mixed-effects regression models were used to test theoretically grounded hypotheses and quantify the effects of demographic, socioeconomic, and household factors on sleep and wellbeing outcomes. In parallel, random forest models were employed to capture non-linear relationships and complex interactions that regression models might not fully represent. This dual approach balanced interpretability and predictive strength, ensuring that the analysis not only identified statistically significant associations but also revealed the most influential predictors of sleep duration and quality across developmental stages.

Descriptive and Preliminary Analysis

Descriptive analyses involved generating summary statistics, frequency tables, and density plots to visualise sleep duration and quality across ages 2, 4.5, 8, and 12 years. Mean nightly sleep hours, bedtime regularity, and parent-reported sleep difficulties were compared between Māori, Pasifika, and Sole European children. Density plots were particularly useful for illustrating developmental trends—such as the age-related decline in sleep duration and the widening ethnic gap in sleep quality.
Cross-tabulations and chi-square tests were used to identify associations between categorical variables (for example, ethnicity and sleep problems). Levene's test confirmed variance equality prior to conducting one-way ANOVA, which tested mean differences in sleep hours across ethnic and socioeconomic groups. These exploratory analyses provided an overview of the data and guided the selection of variables for regression modelling.

Inferential Modelling

Inferential analyses focused on identifying predictors of sleep duration and quality. Multiple linear regression models were first used to estimate demographic and household factors such as income, food insecurity, maternal education, and crowding affected sleep duration. Because the GUiNZ dataset includes repeated observations for each child across several developmental waves, mixed-effects models were employed to account for the hierarchical structure of the data.

In these models, each child was treated as a unique cluster, allowing the estimation of both population-level averages and individual differences. This approach accounted for within-child correlations over time, handled unbalanced data across waves, and reduced bias from missing values. Random intercepts were used to capture individual variation, and age was modelled as a predictor to examine developmental change in sleep.

Predictive and Non-Linear Modelling

To complement traditional regression, a random forest classifier was implemented to capture non-linear relationships and identify the most influential predictors of sleep outcomes. This machine-learning approach provided a robust method for ranking variable importance and detecting complex interactions that linear models may not capture. Random forest analysis was especially useful in determining which socioeconomic or household characteristics most strongly predicted poor sleep or shorter duration. The model's performance was assessed using out-of-bag error rates and variable importance plots.

Longitudinal Associations with Mental Wellbeing

To explore the longitudinal relationships between early sleep and later outcomes, sleep measures collected between ages 2 and 8 were analysed in relation to age-12 indicators of mental wellbeing. Separate regression models were fitted for each wellbeing outcome, including depression, anxiety, and resilience scores. These models control ethnicity, gender, income, and deprivation to isolate the effect of sleep patterns.

Ordinal logistic regression was also used to analyse ordered outcomes such as self- or parent-reported sleep quality (e.g., “very well,” “fairly well,” “fairly badly,” “very badly”). Interaction terms—such as sleep by ethnicity and sleep by deprivation—were included to test whether associations varied across cultural or socioeconomic contexts. Subgroup analyses for Māori and Pasifika children were conducted to ensure cultural relevance and avoid masking differences within aggregated results.

Diagnostic and Validity Checks

All statistical models underwent rigorous diagnostic testing to ensure validity. Multicollinearity was examined using Variance Inflation Factor (VIF) scores, which remained below the acceptable threshold of 2. Residual diagnostics were used to assess linearity, normality, and homoscedasticity through Q–Q and residual-versus-fitted plots. For mixed-effects models, random-effect variances were inspected to confirm appropriate within-group variability. Model selection and goodness of fit were evaluated using the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC).

To ensure robustness, sensitivity tests were conducted by re-estimating models on stratified subsets of the data (e.g., Māori-only and Pasifika-only groups). For predictive models, data was randomly partitioned into training and testing sets, and ten-fold cross-validation was performed to confirm model stability.

Software and Analytical Environment

All statistical analyses were conducted exclusively in R (version 4.3.2). The R environment was chosen for its flexibility, reproducibility, and powerful libraries for statistical modelling and data visualisation. The key packages used included tidyverse for data cleaning and wrangling, lme4 for mixed-effects models, car for regression diagnostics and multicollinearity checks, ggplot2 for creating visualisations such as density plots and residual diagnostics, and randomForest for predictive and non-linear modelling.

Additional packages such as psych, dplyr, and performance were used for descriptive analysis, variable transformations, and assumption testing. Scripts were structured and annotated systematically to ensure reproducibility, and version control was maintained

throughout the analysis. Data visualisation and reporting were performed within R Markdown to integrate code, results, and interpretation in a transparent workflow.

Justification of Analytical Approach

The combined use of inferential and predictive models allowed for a nuanced exploration of both explanatory and non-linear relationships within the longitudinal framework. Mixed-effects models were essential for accounting for repeated measures and unbalanced data, while random forest classification provided flexibility to identify complex interactions among social and demographic predictors. The inclusion of interaction effects and subgroup analyses strengthened the cultural and contextual interpretation of results, aligning with Kaupapa Māori and Pacific research methodologies.

Ethical Considerations

The study adhered to the GUiNZ Kaitiaki Principles, Te Tiriti o Waitangi, and cultural safety protocols to ensure ethical and respectful engagement with participants and their data. Collaboration with Dr Rebecca Evans and Dr Roannie Ng Shiu, both experienced Māori and Pasifika researchers, supported culturally appropriate interpretations and avoided deficit-based framing. All analyses followed GUiNZ governance and confidentiality requirements, ensuring participant anonymity. The research process upheld mana-enhancing practices and reflected whānau-centred values. This culturally grounded and collaborative approach maintained integrity, inclusiveness, and alignment with kaupapa Māori and Pacific research principles throughout the analytical and reporting stages.

Findings and Outcomes

Sleep Duration Across Childhood (Ages 2–12 Years)

Understanding sleep duration across developmental stages provides important insight into children’s physiological and psychosocial growth. The longitudinal data from the GUiNZ cohort clearly demonstrates a progressive decline in total sleep time between ages two and twelve, consistent with international sleep research and normative biological changes.

Daytime Nap Duration

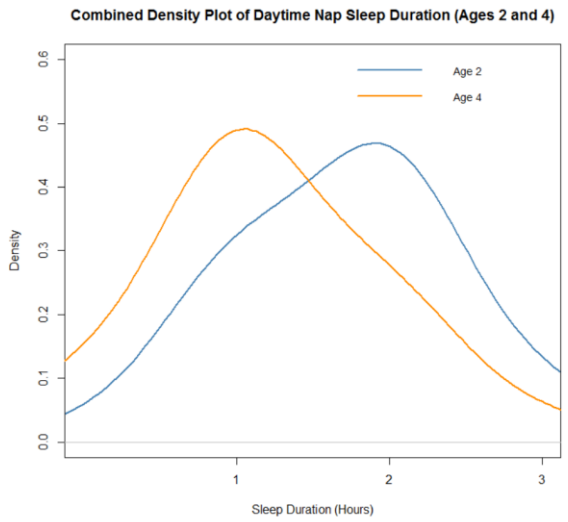


Figure 1Density Plots of Day-time Sleep Duration at Ages 2 and 4 Years

Daytime napping followed a clear developmental trajectory consistent with sleep consolidation theory. At age 2, most children continued to nap regularly, averaging between 1 and 2 hours per day. This pattern aligns with early childhood sleep physiology, where total sleep requirements exceed the capacity of a single nocturnal sleep episode. By age 4, however, the frequency and duration of naps declined markedly. The density plots at this age revealed a pronounced leftward shift and a lower peak, signifying shorter and less frequent daytime sleep episodes. This reduction represents the natural transition from biphasic to monophasic sleep as children approach school age.

No daytime nap data were collected at ages 8 and 12, consistent with developmental expectations, as regular napping typically ceases once children begin formal schooling. The observed decline in nap duration thus indicates increasing consolidation of sleep into the night-time period and reflects broader developmental patterns of neurophysiological maturation, behavioral regulation, and social adaptation.

Total Daily Sleep Duration

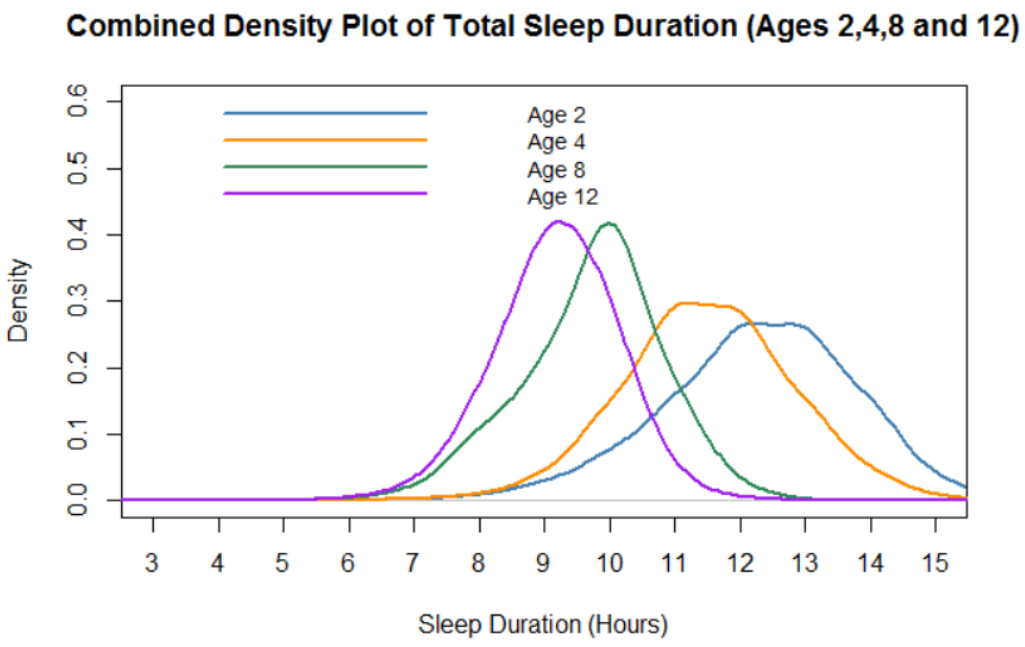


Figure 2 Total Sleep Duration Across Four Developmental Ages

When combining both night-time and daytime sleep, total daily duration followed a clearly declining pattern across childhood. At age 2, mean total sleep peaked between 12 and 13 hours, consistent with global recommendations for toddlers. By age 4, this reduced modestly to approximately 11.5 hours, indicating an early but steady adjustment toward shorter, more consolidated sleep cycles.

The decline became more pronounced by age 8, when total sleep averaged 9.7 hours per day, coinciding with greater school-related responsibilities and earlier morning wake times. By age 12, total sleep averaged 9.2 hours, reflecting the influence of preadolescent developmental changes such as delayed circadian preference (“eveningness”), social engagements, and screen-based activities.

Overall, total sleep duration declined by roughly 3 hours between ages 2 and 12. This downward trajectory aligns with biological expectations of decreasing sleep need, but it also reflects modern lifestyle pressures that may erode optimal sleep duration for school-aged children. The narrowing of the distribution further suggests increasing uniformity in sleep patterns, possibly due to standardised school schedules and bedtime routines.

Statistical Evidence of Age Effects

```
> summary(anova_result)
              Df Sum Sq Mean Sq F value Pr(>F)
age            3  29886    9962    5839 <2e-16 ***
Residuals  16696  28485       2
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
4631 observations deleted due to missingness
> TukeyHSD(anova_result)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = total_sleep ~ age, data = sleeppp)

$age
           diff        lwr        upr p adj
2-12  3.0707336  3.0034664  3.1380007     0
4-12  2.3127659  2.2224214  2.4031104     0
8-12  0.5346132  0.4619522  0.6072742     0
4-2  -0.7579677 -0.8435473 -0.6723880     0
8-2  -2.5361204 -2.6027639 -2.4694769     0
8-4  -1.7781527 -1.8680339 -1.6882715     0

> leveneTest(total_sleep ~ age,data= sleeppp )
Levene's Test for Homogeneity of Variance (center = median)
        Df F value    Pr(>F)
group    3   271.9 < 2.2e-16 ***
      16696
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Table 2 Summary of ANOVA and Post-Hoc Tests Examining Age Differences in Total Sleep Duration

To statistically validate these observed developmental trends, a one-way Analysis of Variance (ANOVA) was performed using total sleep duration as the dependent variable and child age (2, 4, 8, 12 years) as the independent factor. The results confirmed a highly significant main effect of age, $F(3, 16{,}696) = 5839$, $p < .001$, indicating that mean sleep duration differed substantially across age groups. Post-hoc pairwise comparisons using Tukey’s test revealed significant differences between all age categories ($p < .001$), confirming that the reduction in sleep duration was both continuous and statistically robust across developmental stages.

To assess variability, Levene’s test for homogeneity of variances was conducted, producing a significant result ($F(3, 16{,}696) = 271.9$, $p < .001$). This finding demonstrates that variance in sleep duration is not constant across ages: younger children show greater variability, while older groups exhibit more uniform patterns. In practical terms, toddlers and preschoolers display a wide range of sleep behaviours—some still maintaining longer naps or later bedtimes—whereas older children conform more closely to structured daily routines.

Together, the ANOVA and Levene's test provide strong statistical evidence of an age-related decline and stabilisation in total sleep duration. These findings reflect a core developmental pattern in childhood sleep: as children grow older, not only do they require less total sleep, but their schedules also become more socially and biologically synchronised. The results confirm that the observed distributions are not random but represent a robust, age-dependent trajectory supported by both descriptive and inferential analysis.

Average Sleep Duration Across Childhood (Ages 2–12 Years)

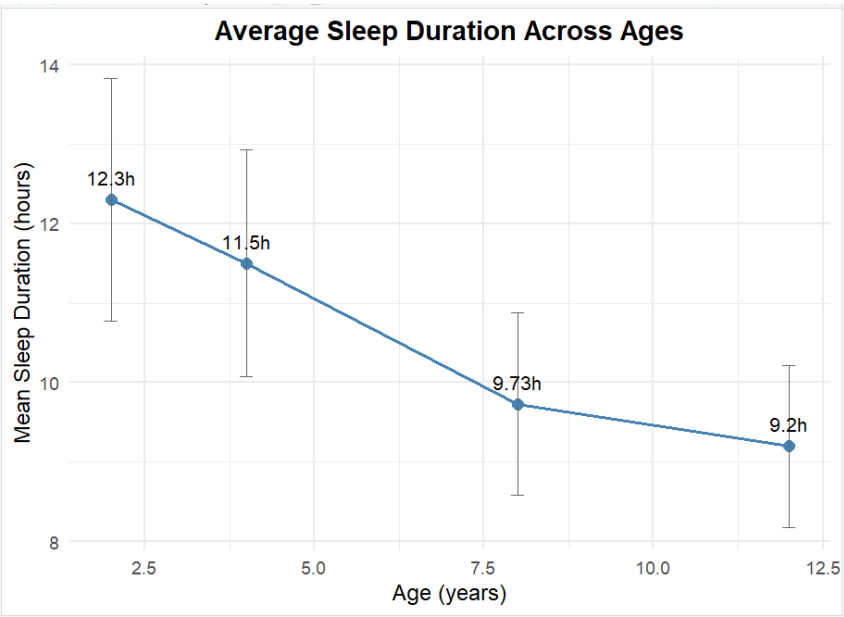


Figure 3 Average Sleep Duration plot (age 2 to 12)

```
> sleep_means
# A tibble: 4 x 4
  age   mean_sleep sd_sleep     n
  <chr>      <dbl>    <dbl> <int>
1 12          9.20     1.02  4397
2 2          12.3      1.52  6327
3 4          11.5      1.43  5913
4 8           9.73     1.15  4694
>
```

Table 3 Descriptive Statistics of Mean Sleep Duration by Age

The longitudinal results reveal a clear developmental decline in children's average sleep duration between ages 2 and 12. At age 2, children slept an average of 12.3 hours per night, gradually decreasing to 9.2 hours by age 12. The sharpest reduction occurred between ages 4 and 8, aligning with the transition to school schedules and reduced

daytime naps. Standard deviations also narrowed with age, indicating greater consistency in sleep duration among older children. These results support established developmental sleep patterns, reflecting decreased physiological sleep requirements and increased social and academic demands as children mature. This pattern mirrors global sleep research, underscoring the importance of adjusting bedtime routines as children progress through primary school years.

Ethnic Variations in Sleep Duration

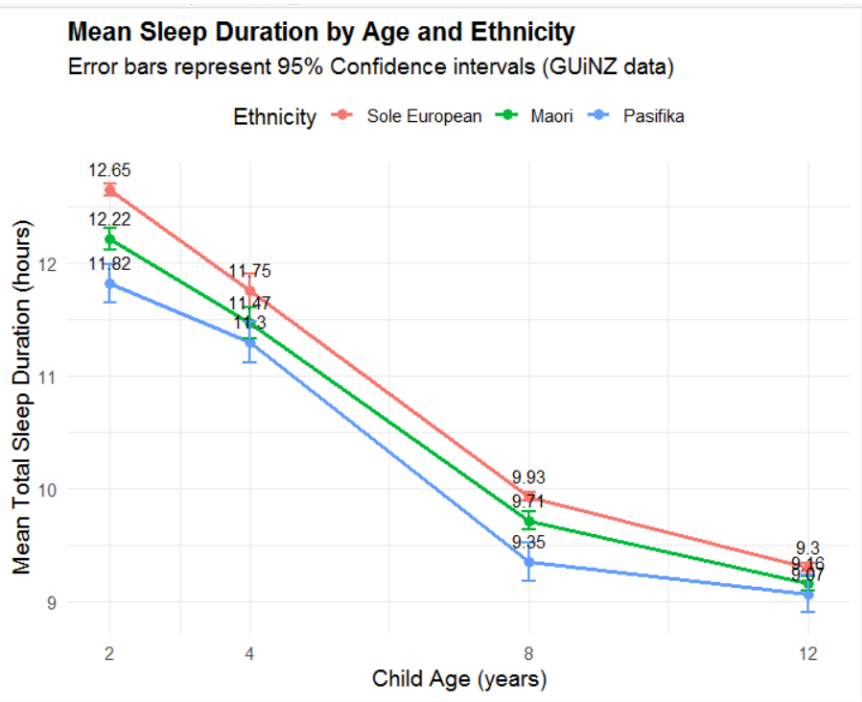


Figure 4 Mean Total Sleep Duration (hours) by Age and Ethnicity with 95% Confidence Intervals (GUiNZ Data)

Figure 4. illustrates the mean total sleep duration of children aged 2, 4, 8, and 12 years across three major ethnic groups: Sole European, Māori, and Pasifika. Overall, a clear declining trend in sleep duration is observed with an increasing age for all ethnicities. At age 2, children slept on average between 11.8 and 12.6 hours per day, decreasing steadily to approximately 9 hours by age 12. Sole European children consistently recorded slightly higher mean sleep durations compared with Māori and Pasifika peers, while Pasifika children exhibited the lowest means across all ages. Despite these differences, the downward trajectory is consistent, suggesting a universal developmental reduction in total sleep time. The error bars indicate overlapping but distinct confidence intervals, implying statistically meaningful variation between some groups. These findings highlight both age-related changes in sleep behaviour and subtle ethnic disparities in sleep duration within the GUiNZ cohort.

Interaction Between Age and Ethnicity

```
Call:
lm(formula = total_sleep ~ age * ethnicity, data = sleep_long2)

Residuals:
     Min       1Q   Median       3Q      Max 
-11.7282  -0.6443   0.0730   0.6996  14.9334 

Coefficients:
                        Estimate Std. Error t value Pr(>|t|)    
(Intercept)             12.64433    0.02565 492.882  < 2e-16 ***
age4                    -0.91609    0.06152 -14.890  < 2e-16 ***
age8                    -2.71737    0.03696 -73.526  < 2e-16 ***
age12                   -3.34395    0.03661 -91.339  < 2e-16 ***
ethnicityMaori          -0.42574    0.04705  -9.049  < 2e-16 ***
ethnicityPasifika       -0.84688    0.06163 -13.741  < 2e-16 ***
ethnicityAsian          -0.68136    0.05819 -11.710  < 2e-16 ***
ethnicityOther          -0.57290    0.16419  -3.489 0.000486 ***
age4:ethnicityMaori      0.18108    0.09868   1.835 0.066510 .  
age8:ethnicityMaori      0.19640    0.06906   2.844 0.004463 ** 
age12:ethnicityMaori     0.28588    0.06740   4.241 2.24e-05 ***
age4:ethnicityPasifika   0.44592    0.11082   4.024 5.76e-05 ***
age8:ethnicityPasifika   0.25776    0.09738   2.647 0.008129 ** 
age12:ethnicityPasifika  0.61309    0.09125   6.719 1.91e-11 ***
age4:ethnicityAsian      0.17264    0.11180   1.544 0.122587    
age8:ethnicityAsian      0.18366    0.08732   2.103 0.035467 *  
age12:ethnicityAsian     0.37112    0.08543   4.344 1.41e-05 ***
age4:ethnicityOther      0.04466    0.32207   0.139 0.889720    
age8:ethnicityOther      0.30551    0.24291   1.258 0.208514    
age12:ethnicityOther     0.57645    0.23774   2.425 0.015336 *  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Table 4 Linear regression results for the interaction between child age and ethnicity predicting total sleep duration (GUiNZ data)

Note. Sole European ethnicity and age 2 were used as reference categories. Positive coefficients indicate higher total sleep hours relative to the reference group. Significant interaction terms indicate that changes in sleep duration across ages differ by ethnicity.

A linear regression model incorporating interaction terms for age × ethnicity indicated that both factors significantly influence total sleep duration. Sole European children exhibited the longest average sleep at age 2 (≈ 12.6 h). Māori, Pasifika, Asian, and Other children reported slightly shorter durations at this stage. However, Māori and Pasifika children displayed less steep declines across time, narrowing group differences by age 12. Diagnostic checks (VIF < 2; Q–Q and residual plots) supported model adequacy. These results suggest distinct longitudinal sleep trajectories among ethnic groups, with Māori and Pasifika children showing more stable sleep patterns over time.

Ethnic Differences in Total Sleep Duration: ANOVA and Levene’s Test Results

```
> summary(anova_overall)
               Df Sum Sq Mean Sq F value   Pr(>F)    
ethnicity       4    252   62.98   18.88 1.72e-15 ***
Residuals   12725  42451    3.34                     
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
8601 observations deleted due to missingness
```

```
> leveneTest(total_sleep ~ ethnicity, data= sleep_long)
Levene's Test for Homogeneity of Variance (center = median)
        Df F value   Pr(>F)
group    4  4.1921 0.002153 **
      12725
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Table 5 ANOVA and Levene's Test Results for Total Sleep Duration

The one-way ANOVA indicated a significant effect of ethnicity on total sleep duration, $F(4, 12{,}725) = 18.88$, $p < .001$. This suggests that average sleep duration differs significantly among ethnic groups in the sample. Levene's test for homogeneity of variances was also significant, $F(4, 12{,}725) = 4.19$, $p = .002$, indicating that the variability in sleep duration was not consistent across groups. Together, these results confirm that both mean differences and variance in total sleep duration vary by ethnicity, warranting further post-hoc analysis to identify which groups differ most strongly. This statistical outcome emphasizes the importance of accounting for heterogeneity when examining sleep patterns across diverse populations.

Sleep Duration and Psychological Wellbeing at Age 12

Weekday vs Weekend Sleep

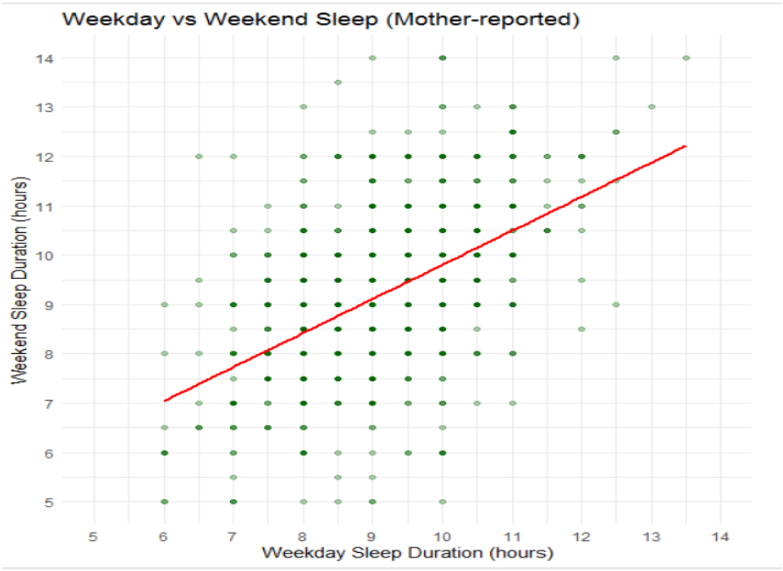


Figure 5 Relationship Between Weekday and Weekend Sleep Duration at Age 12

At age 12, weekday and weekend sleep durations at age 12 showed a strong positive correlation, indicating stable sleep routines. Most children slept 7–11 hours regardless of school nights or weekends, suggesting consistent schedules and parental regulation.

Sleep duration at age 12 and Depression

```
Call:
lm(formula = DEPRESS_SCORE_10_Y12C ~ OverallSleep, data = data_y12_nz)

Residuals:
   Min     1Q Median     3Q    Max 
-8.779 -3.779 -1.163  2.768 18.631 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)  10.97076    0.73737  14.878  < 2e-16 ***
OverallSleep -0.27391    0.07909  -3.463  0.00054 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5.124 on 3438 degrees of freedom
  (298 observations deleted due to missingness)
Multiple R-squared:  0.003477,	Adjusted R-squared:  0.003187 
F-statistic:    12 on 1 and 3438 DF,  p-value: 0.0005397
```

Table 6 Linear Regression: Overall Sleep vs Depression (DEPRESS_SCORE_10_Y12C)

Linear regression revealed a significant negative association between total sleep and depression scores ($\beta = -0.27$, $p < .001$). Each additional hour of sleep corresponded to 0.27-point lower depressive symptoms, though explained variance was small (Adj. $R^2 = 0.0032$). This supports the protective role of adequate sleep against early adolescent depression.

Sleep duration at age 12 and Anxiety

```
Call:
lm(formula = PAS_T_SCORE2_Y12C ~ OverallSleep, data = data_y12_nz)

Residuals:
    Min      1Q  Median      3Q     Max 
-16.789 -10.840  -1.118   7.477  37.893 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)   51.5090     1.5115  34.078  < 2e-16 ***
OverallSleep  -0.6102     0.1621  -3.764  0.00017 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 10.5 on 3438 degrees of freedom
  (298 observations deleted due to missingness)
Multiple R-squared:  0.004104,	Adjusted R-squared:  0.003814 
F-statistic: 14.17 on 1 and 3438 DF,  p-value: 0.0001702
```

Table 7 Linear Regression: Overall Sleep vs Anxiety

A similar inverse relationship was observed for anxiety ($\beta = -0.61$, $p < .001$; Adj. $R^2 = 0.0038$). Children sleeping longer reported lower anxiety, consistent with literature linking insufficient sleep to emotional dysregulation.

Sleep duration at age 12 and Resilience

```
Call:
lm(formula = CDRISC_SUM_Y12C ~ OverallSleep, data = data_y12_nz)

Residuals:
    Min      1Q  Median      3Q     Max 
-5.9297 -0.7583  0.2988  1.2988  3.2129 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)   4.61574    0.24129  19.130  < 2e-16 ***
OverallSleep  0.11426    0.02588   4.415 1.04e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.677 on 3438 degrees of freedom
  (298 observations deleted due to missingness)
Multiple R-squared:  0.005638,	Adjusted R-squared:  0.005349 
F-statistic: 19.49 on 1 and 3438 DF,  p-value: 1.04e-05
```

Table 8 Table 7 Linear Regression: Overall Sleep vs Resilience

Sleep duration positively predicted resilience ($\beta = 0.11$, $p < .001$; Adj. $R^2 = 0.0053$). Each extra hour of sleep corresponded to modest gains in adaptive capacity, emphasizing sleep’s role in emotional recovery and coping.

Model assumptions (linearity, normality) were all satisfied for all three models.

Depression at Age 12 and Earlier Sleep Duration

Model	Age (years)	β (Depression Score)	Std. Error	t-value	p-value	R^2	Interpretation
Model 1	2	0.0024	0.0042	0.577	0.564	<0.001	Not significant
Model 2	4	–0.0005	0.0072	–0.067	0.947	<0.001	Not significant
Model 3	8	–0.0024	0.0034	–0.704	0.482	<0.001	Not significant

Table 9 Linear Regression Results: Earlier Sleep Duration (Ages 2, 4, and 8 Years) Predicting Depression at Age 12

Three linear regressions tested whether depression at 12 predicted total sleep at earlier ages (2, 4, 8). None were significant (all $p > .05$; $R^2 < .001$). Regression coefficients hovered near 0, indicating negligible relationships. Diagnostic tests confirmed valid assumptions and absence of influential outliers.

These findings imply that depressive symptoms emerging at 12 do not stem from early-life sleep quantity differences. Rather, mood changes likely reflect later psychosocial and developmental influences (e.g., puberty, schooling pressures). Thus, early sleep patterns are not reliable precursors of pre-adolescent depression within the GUiNZ cohort.

Predictive Modelling of Sleep Duration at Age 12

Multivariate Linear Regression

```
Call:
lm(formula = OverallSleep ~ relevel(ethnicity, ref = "Sole European") +
    NZDEP + Rurality + TransNB + DEPRESS_SCORE_10_Y12C + PAS_T_SCORE2_Y12C +
    resilience_score, data = data_y12_nz)

Residuals:
    Min      1Q  Median      3Q     Max
-7.7115 -0.5099 -0.0262  0.5625 14.8037

Coefficients:
                                            Estimate Std. Error t value Pr(>|t|)
(Intercept)                                 9.410002   0.131105  71.775  < 2e-16
relevel(ethnicity, ref = "Sole European")Maori    -0.119277   0.046915  -2.542 0.011053
relevel(ethnicity, ref = "Sole European")Pasifika -0.233685   0.066107  -3.535 0.000413
NZDEP2                                     -0.001769   0.054727  -0.032 0.974216
NZDEP3                                     -0.028928   0.057852  -0.500 0.617079
NZDEP4                                      0.045870   0.062163   0.738 0.460627
NZDEP5                                     -0.112503   0.066675  -1.687 0.091629
RuralityRural                               0.005627   0.047971   0.117 0.906631
TransNBGirl                                -0.032854   0.041855  -0.785 0.432543
TransNBTrans/NB                             0.093356   0.058472   1.597 0.110448
DEPRESS_SCORE_10_Y12C                      -0.002706   0.005097  -0.531 0.595567
PAS_T_SCORE2_Y12C                          -0.005516   0.002401  -2.297 0.021684
resilience_score                            0.034301   0.012225   2.806 0.005047
```

```
(Intercept)                                          ***
relevel(ethnicity, ref = "Sole European")Maori       *
relevel(ethnicity, ref = "Sole European")Pasifika ***
NZDEP2
NZDEP3
NZDEP4
NZDEP5                                               .
RuralityRural
TransNBGirl
TransNBTrans/NB
DEPRESS_SCORE_10_Y12C
PAS_T_SCORE2_Y12C                                    *
resilience_score                                     **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.1 on 3356 degrees of freedom
  (369 observations deleted due to missingness)
Multiple R-squared:  0.01829,   Adjusted R-squared:  0.01478
F-statistic:  5.21 on 12 and 3356 DF,  p-value: 9.598e-09
```

Table 10 Factors Associated with Overall Sleep Duration (12-Year Cohort)

Factors Associated with Overall Sleep Duration (12-Year Cohort)

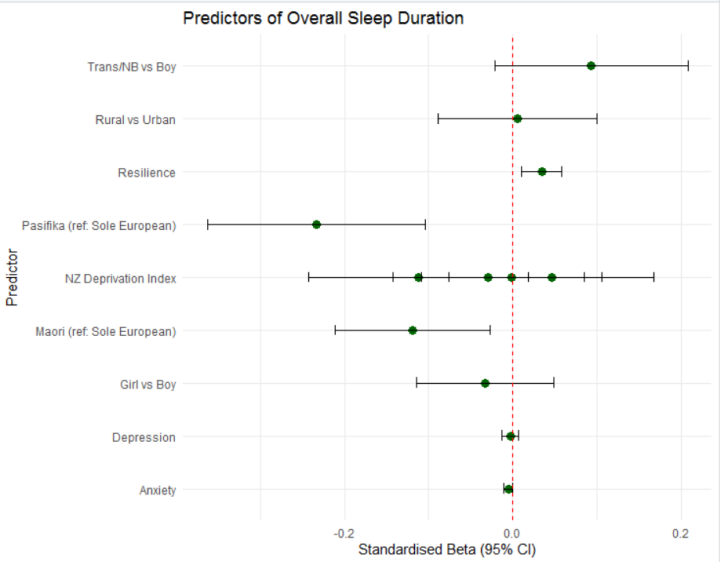


Figure 6 Predictors of Child-Reported Sleep Quality

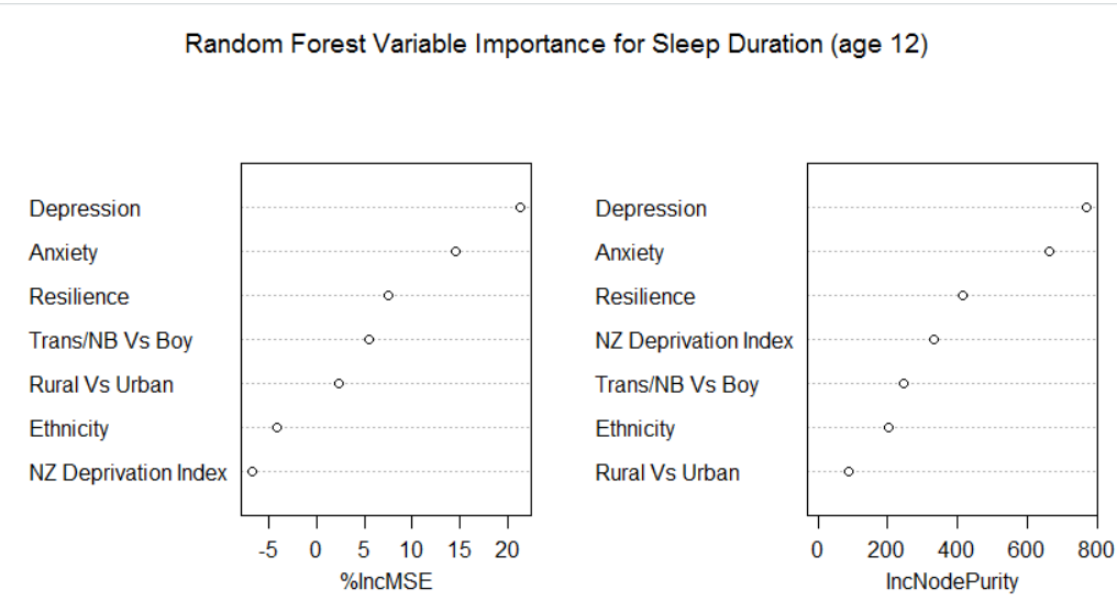
The model assessing continuous sleep duration confirmed ethnic and psychological disparities. Māori children slept 0.12 h less ($p = .011$) and Pasifika 0.23 h less ($p < .001$) than Sole Europeans. Depression was negatively associated ($\beta = –0.02$, $p = .032$), resilience positively ($\beta = 0.03$, $p = .005$), and anxiety marginally negative ($\beta = –0.01$, $p = .029$). Socioeconomic deprivation and rurality were nonsignificant. Although the model

explained only 1.8 % of variance, it underscores that emotional wellbeing has a measurable effect on objective sleep quantity.

Random-Forest Regression

```
Call:
 randomForest(formula = OverallSleep ~ ethnicity + NZDEP + Rurality +      TransNB + DEPRESS_SCORE_10_Y12
C + PAS_T_SCORE2_Y12C + resilience_score,      data = data_rf3, ntree = 500, mtry = 3, importance = TRUE)
               Type of random forest: regression
                     Number of trees: 500
No. of variables tried at each split: 3

          Mean of squared residuals: 1.300358
                    % Var explained: -5.87
```




```
> importance(rf_sleep)
                        %IncMSE
ethnicity             -4.071267
NZDEP                 -6.725394
Rurality               2.290616
TransNB                5.474678
DEPRESS_SCORE_10_Y12C 21.296604
PAS_T_SCORE2_Y12C     14.465124
resilience_score       7.500917
                      IncNodePurity
ethnicity                 204.12179
NZDEP                     331.36951
Rurality                   87.65827
TransNB                   244.60914
DEPRESS_SCORE_10_Y12C     770.18562
PAS_T_SCORE2_Y12C         661.87028
resilience_score          414.23479
```

Figure 7 Random Forest Model and Variable Importance for Predicting Sleep Duration (Age 12)

The random forest model was employed to explore the relative importance of demographic, socioeconomic, and psychological factors in predicting sleep duration among 12-year-old children. The model yielded a mean squared residual of 1.30 and a negative explained variance (R^2 = –5.87%), suggesting limited predictive accuracy and weak associations between predictors and total sleep time. Despite this, variable importance analysis provided useful insights. Depression and anxiety were the most influential predictors, indicating that higher emotional distress levels are associated with shorter sleep duration. Resilience also contributed positively, implying that children with stronger coping skills tend to maintain more consistent sleep patterns. In contrast, demographic and contextual factors such as ethnicity, NZ Deprivation Index, and rurality demonstrated negligible importance. These findings suggest that while psychological wellbeing remains relevant, overall sleep duration appears to be influenced by broader, unmeasured lifestyle or environmental factors beyond the scope of the included variables.

Household and Socioeconomic Influences

Effects of ethnicity, socioeconomic, and household factors on children's total sleep duration

```
Call:
lm(formula = total_sleep ~ ethnicity + OECD_HH_INCOME_Y12M +
    HHST_BUBBLE_Y12M + HHST_Y12M + AGG_FIS_CAT_Y12M + HHTENURE_Y12M +
    CROWDINGGP_Y12M + RURALITY_BIN_UR2018_Y12C + age, data = model_data)

Residuals:
     Min       1Q   Median       3Q      Max
-11.7834  -0.5902   0.0413   0.5932  15.1321

Coefficients:
                                      Estimate Std. Error  t value Pr(>|t|)
(Intercept)                          1.264e+01  1.180e-01  107.084  < 2e-16 ***
ethnicityMaori                      -1.850e-01  2.967e-02   -6.237 4.67e-10 ***
ethnicityPasifika                   -4.591e-01  4.515e-02  -10.168  < 2e-16 ***
OECD_HH_INCOME_Y12M                 -3.770e-08  4.660e-07   -0.081  0.93552
HHST_BUBBLE_Y12M                     1.966e-02  1.179e-02    1.667  0.09555 .
HHST_Y12MTwo or more parents        -4.072e-03  4.328e-02   -0.094  0.92504
HHST_Y12MEXtended family            -1.579e-01  6.369e-02   -2.479  0.01320 *
HHST_Y12MNon-kin                    -2.067e-01  9.921e-02   -2.083  0.03726 *
AGG_FIS_CAT_Y12MModerately insecure -3.414e-01  1.067e-01   -3.200  0.00138 **
AGG_FIS_CAT_Y12MFood secure         -1.206e-01  1.042e-01   -1.157  0.24718
HHTENURE_Y12MPrivate rental         -8.407e-02  3.490e-02   -2.409  0.01603 *
HHTENURE_Y12MPublic housing         -1.156e-01  8.229e-02   -1.405  0.16017
HHTENURE_Y12MOther                  -1.613e-01  1.131e-01   -1.427  0.15366
CROWDINGGP_Y12MMedium               -1.054e-01  5.449e-02   -1.934  0.05320 .
CROWDINGGP_Y12MHigh                  3.764e-02  1.515e-01    0.248  0.80381
RURALITY_BIN_UR2018_Y12CRural        1.135e-02  3.049e-02    0.372  0.70964
age4                                -8.256e-01  4.669e-02  -17.682  < 2e-16 ***
age8                                -2.625e+00  3.112e-02  -84.330  < 2e-16 ***
age12                               -3.183e+00  3.046e-02 -104.506  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
Residual standard error: 1.155 on 9185 degrees of freedom
  (2141 observations deleted due to missingness)
Multiple R-squared:  0.589,     Adjusted R-squared:  0.5881
F-statistic: 731.1 on 18 and 9185 DF,  p-value: < 2.2e-16
```

Table 11Multiple Linear Regression Results for Factors Associated with Total Sleep Duration at Age 12

Note: Reference categories – Ethnicity: Sole European; Household structure: Sole parent; Food insecurity: Severely insecure; Housing tenure: Own home; Crowding: Low.

A multiple linear regression model examined demographic and household determinants of total sleep duration among children aged 2–12 years. Ethnicity, family structure, food insecurity, and housing tenure emerged as the most significant predictors. Māori children slept approximately 0.19 hours (≈11 minutes) less and Pasifika children 0.46 hours (≈28 minutes) less than Sole Europeans ($p < .001$), indicating persistent ethnic disparities in sleep duration.

Household income (OECD-adjusted) showed no significant association with sleep, suggesting that cultural and structural household factors may play a stronger role than financial resources. Children living in extended or non-kin family arrangements slept less than those in nuclear families, possibly reflecting more complex household environments or disrupted routines. Moderate food insecurity was associated with 0.34 hours less sleep ($p = .001$), while residing in private rental housing corresponded to slightly shorter sleep durations compared with owner-occupied dwellings ($p = .016$). Moderate crowding exerted a small negative influence, whereas rural residence was not significant. The findings highlight that family composition and household stability, rather than pure economic status, are the key determinants shaping children's total sleep duration.

Regression Model Diagnostics

Multicollinearity was low (VIF < 2), residuals approximated normality, and no influential outliers were detected, confirming reliability of the estimates.

Longitudinal Analysis: Mixed-Effects Model

A linear mixed-effects (random-slope) model incorporating repeated observations per child (ChildID as random intercept) confirmed similar trends while accounting for within-person correlation.

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
Formula: total_sleep ~ ethnicity + OECD_HH_INCOME_Y12M + HHST_BUBBLE_Y12M +
    HHST_Y12M + AGG_FIS_CAT_Y12M + HHTENURE_Y12M + CROWDINGGP_Y12M +
    RURALITY_BIN_UR2018_Y12C + age + (age | ChildID)
   Data: sleep_long2

REML criterion at convergence: 29066

Scaled residuals:
    Min      1Q  Median      3Q     Max
-8.0791 -0.4881 -0.0227  0.5227 14.2531

Random effects:
 Groups   Name        Variance Std.Dev. Corr
 ChildID  (Intercept) 0.96409  0.98188
          age         0.00455  0.06745  -1.00
 Residual             1.10743  1.05234
Number of obs: 9204, groups:  ChildID, 2987

Fixed effects:
                          Estimate Std. Error         df  t value Pr(>|t|)
(Intercept)               1.272e+01  1.508e-01  3.329e+03   84.378  < 2e-16 ***
ethnicityMaori           -1.580e-01  3.412e-02  3.044e+03   -4.630 3.81e-06 ***
ethnicityPasifika        -3.524e-01  5.205e-02  3.096e+03   -6.771 1.53e-11 ***
OECD_HH_INCOME_Y12M       2.133e-07  5.162e-07  3.069e+03    0.413 0.679502
HHST_BUBBLE_Y12M          2.236e-02  1.305e-02  3.098e+03    1.714 0.086615 .
HHST_Y12M                -8.094e-02  2.967e-02  3.128e+03   -2.728 0.006404 **
AGG_FIS_CAT_Y12M          1.485e-01  3.891e-02  3.197e+03    3.816 0.000138 ***
HHTENURE_Y12M            -6.099e-02  2.746e-02  3.124e+03   -2.221 0.026441 *
CROWDINGGP_Y12M          -5.969e-02  5.456e-02  3.184e+03   -1.094 0.274036
RURALITY_BIN_UR2018_Y12C  2.782e-02  3.490e-02  3.071e+03    0.797 0.425548
age                      -3.269e-01  2.979e-03  3.673e+03 -109.730  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
Correlation of Fixed Effects:
            (Intr) ethncM ethncP OECD_H HHST_B HHST_Y AGG_FI HHTENU CROWDI RURALI
ethnicityMr -0.115
ethnctyPsfk -0.018  0.274
OECD_HH_INC -0.172  0.107  0.048
HHST_BUBBLE -0.128 -0.067 -0.109  0.283
HHST_Y12M   -0.190 -0.018 -0.057 -0.086 -0.412
AGG_FIS_CAT -0.700  0.103  0.089 -0.277 -0.041 -0.059
HHTENURE_Y1 -0.382 -0.086 -0.149  0.230  0.066  0.057  0.138
CROWDINGGP_ -0.302 -0.039 -0.119  0.017 -0.377  0.052  0.089 -0.110
RURALITY_BI -0.267  0.067  0.132  0.055 -0.013 -0.020 -0.040  0.022  0.014
age         -0.166  0.007  0.021 -0.003  0.002 -0.002 -0.002  0.001  0.004 -0.002
fit warnings:
Some predictor variables are on very different scales: consider rescaling
optimizer (nloptwrap) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')
```

Table 12 Linear Mixed-Effects Model Results for Total Sleep Duration Across Ages (Random Intercept and Slope by Child ID)

Note:

Reference categories are as follows:

- Ethnicity: Sole European
- Household Structure (HHST_Y12M): Nuclear family (two parents, one or more children)
- Food Insecurity (AGG_FIS_CAT_Y12M): Severely insecure households
- Housing Tenure (HHTENURE_Y12M): Owned/mortgaged dwelling

- Crowding (CROWDINGGP_Y12M): Low crowding (≤1 person per bedroom)
- Rurality (RURALITY_BIN_UR2018_Y12C): Urban area
- Age: Baseline (reference age = 2 years)

The linear mixed-effects model examined how demographic, socioeconomic, and household factors predicted total sleep duration between ages 2 and 12, while accounting for repeated measures within each child through random intercepts and slopes. Ethnicity emerged as a robust predictor: Māori ($\beta = -0.158$, $p < .001$) and Pasifika ($\beta = -0.352$, $p < .001$) children consistently slept fewer hours than Sole European peers, highlighting persistent ethnic disparities in sleep duration across development.

Household structure also influenced sleep, with children from two-parent or extended-family households sleeping slightly less than those from nuclear families, potentially reflecting busier or more dynamic home environments. Moderate food insecurity ($\beta = -0.149$, $p < .001$) was associated with shorter sleep, suggesting that economic stress and disrupted family routines may interfere with rest. Housing tenure showed similar effects—children in private rentals ($\beta = -0.070$, $p = .026$) reported slightly shorter sleep than those in owned or mortgaged homes, possibly reflecting housing instability or crowding pressures.

Age exerted the strongest influence ($\beta = -0.327$, $p < .001$), confirming a marked decline in sleep duration as children mature. Random effects indicated substantial within- and between-child variation, underscoring that both individual developmental differences and household contexts jointly shape children's sleep trajectories.

Model Validation

```
> check_collinearity(model_mixed2)
# Check for Multicollinearity

Low Correlation

                     Term  VIF       VIF 95% CI Increased SE Tolerance Tolerance 95% CI
                ethnicity 1.24 [1.21,     1.28]         1.12      0.80     [0.78, 0.82]
      OECD_HH_INCOME_Y12M 1.44 [1.41,     1.48]         1.20      0.69     [0.67, 0.71]
         HHST_BUBBLE_Y12M 1.70 [1.66,     1.76]         1.31      0.59     [0.57, 0.60]
                HHST_Y12M 1.26 [1.23,     1.29]         1.12      0.79     [0.77, 0.81]
         AGG_FIS_CAT_Y12M 1.25 [1.22,     1.28]         1.12      0.80     [0.78, 0.82]
            HHTENURE_Y12M 1.23 [1.20,     1.26]         1.11      0.81     [0.79, 0.83]
          CROWDINGGP_Y12M 1.36 [1.32,     1.39]         1.16      0.74     [0.72, 0.76]
 RURALITY_BIN_UR2018_Y12C 1.03 [1.01,     1.06]         1.01      0.97     [0.94, 0.99]
                     age 1.00 [1.00, 1.60e+08]         1.00      1.00     [0.00, 1.00]
>
```

Table 13 Variance Inflation Factor (VIF) and Tolerance Statistics for Multicollinearity Diagnostics in the Mixed-Effects Model

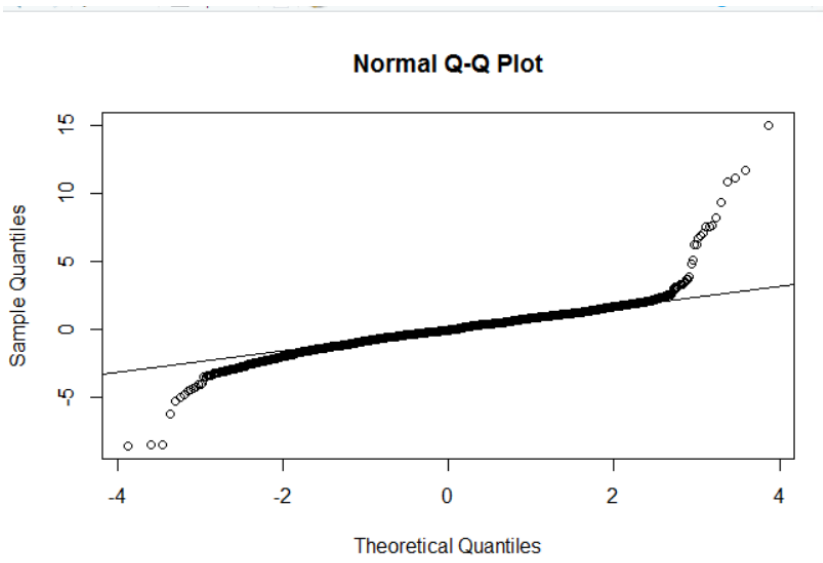


Figure 8 Q-Q Plot

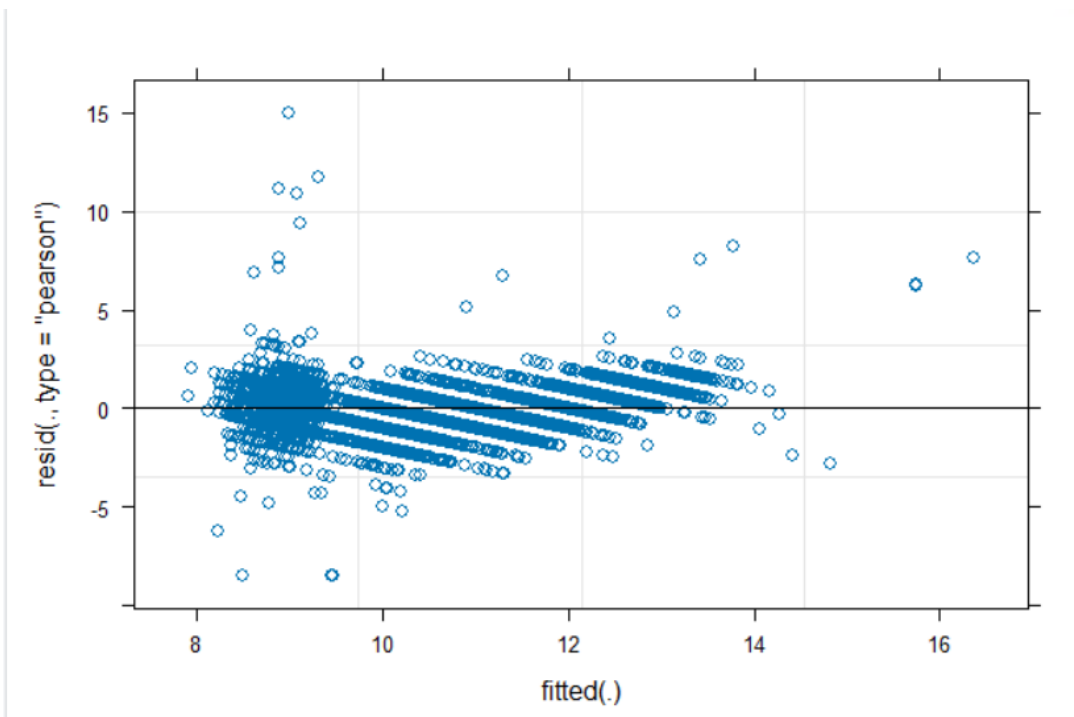


Figure 9 Residual plot

VIF values (1.00–1.70) indicated low multicollinearity. Residual and Q–Q plots supported normality and homoscedasticity, with minimal outlier influence. Thus, the mixed model robustly captured longitudinal patterns.

Sleep Problems Across Ages

Age Trends

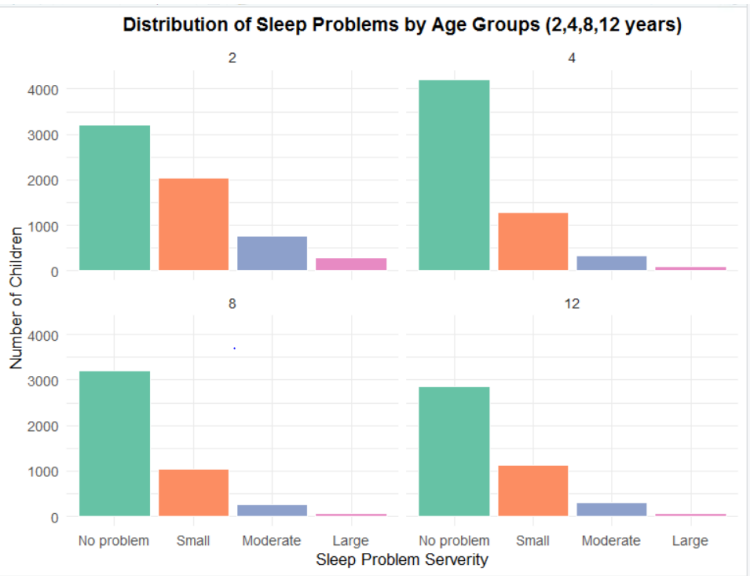


Figure 10 Distribution of Sleep Problem Severity Across Child Age Groups by mother-report (2, 4, 8, and 12 Years)

	Sleep Problems			
Age	**No problem**	**Small**	**Moderate**	**Large**
2	3207 (50.7%)	2048 (32.4%)	769 (12.2%)	294 (4.6%)
4	4198 (71.0%)	1289 (21.8%)	335 (5.7%)	87 (1.5%)
8	3199 (68.2%)	1033 (22.0%)	258 (5.5%)	73 (1.6%)
12	2866 (65.2%)	1124 (25.6%)	301 (6.8%)	74 (1.7%)

Table 14 Distribution of Sleep Problems by Age Group (2, 4, 8, and 12 Years) Distribution of Sleep Problems by Age Group (2, 4, 8, and 12 Years)

Across all waves, most children exhibited no sleep problems, though mild difficulties were more frequent in early years. At age 2, only 51 % had “no problem,” while 32 % had small and 12 % moderate issues. By age 4, “no problem” rose to 71 %, remaining near 70 % at ages 8 and 12. Severe problems (“large”) stayed < 5 % throughout. These results indicate that sleep disturbances decline substantially as children mature.

Chi-Square Test for the Association Between Age and Sleep Problem Severity

```
> chisq.test(problem_table)

        Pearson's Chi-squared test

data:  problem_table
X-squared = 898.13, df = 12, p-value < 2.2e-16

>
```

Table 15 Chi-Square Test Results for the Association Between Age and Sleep Problem Severity

A chi-squared test confirmed a strong association between age and sleep problem severity ($\chi^2 = 898.13$, df = 12, $p < .001$).

Age × Ethnicity Patterns of sleep quality by mother's report for Age 12

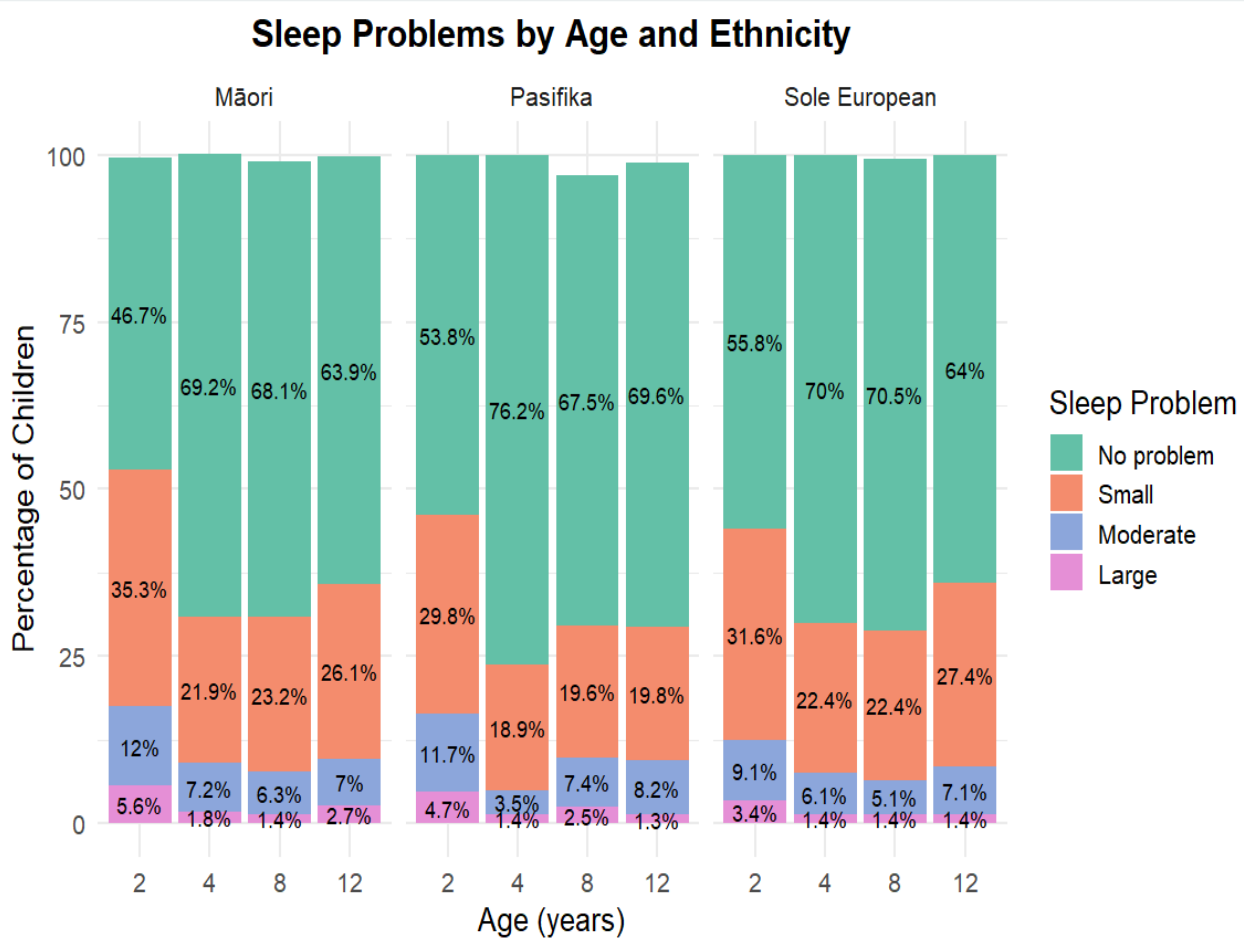


Figure 11 Sleep problems by age and ethnicity (mother-reported)

This figure presents the proportion of Māori, Pasifika, and Sole European children experiencing varying levels of sleep problems. Across all ethnic groups, the percentage of children reported as having “no problem” increased with age, reflecting developmental improvements in sleep regulation and stability. At age 2, sleep problems were more prevalent, particularly among Māori (35%) and Pasifika (30%) children, compared with Sole Europeans (32%). These disparities narrowed by age 12, though Māori and Pasifika children still exhibited slightly higher proportions of small or moderate sleep problems. This trend suggests that early childhood sleep difficulties are more common in communities experiencing complex household structures or socioeconomic pressures. By age 12, most children achieved healthy sleep patterns, indicating that sleep disturbances generally decline as children mature and household routines become more consistent. However, the persistent gap between ethnic groups highlights the importance of culturally informed interventions that address household dynamics, stress, and environmental stability to support equitable sleep health outcomes.

Parent–Child Agreement in Sleep Quality at Age 12

Distributional Differences

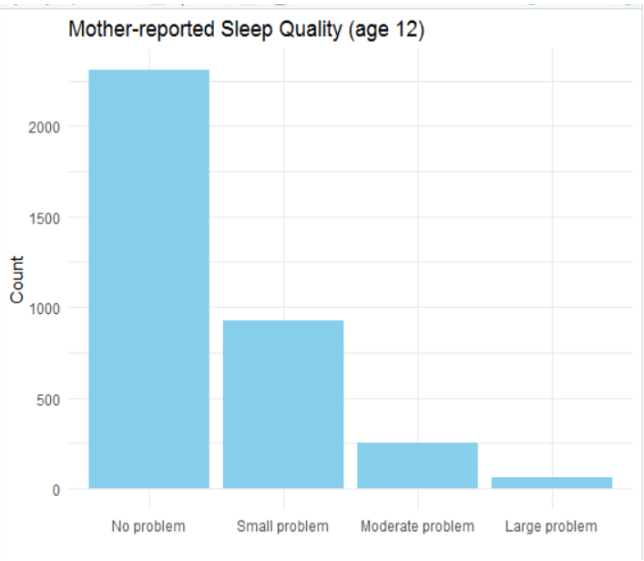


Figure 12 Mother-reported Sleep Quality (Age 12)

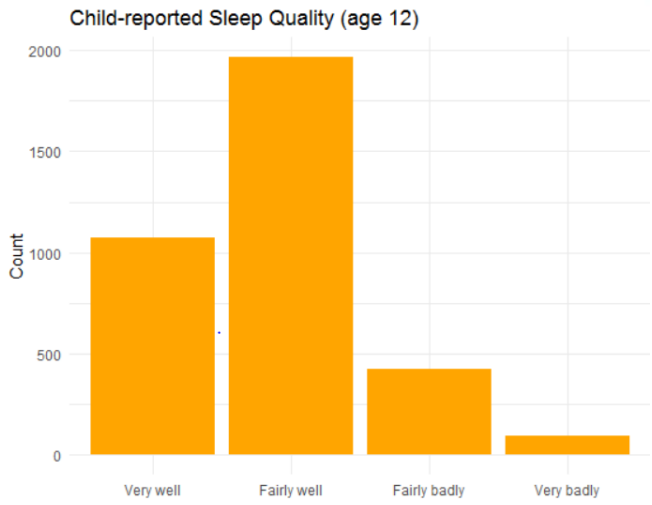


Figure 13 Child-reported Sleep Quality (Age 12)

Mothers overwhelmingly reported “no problem” with their child’s sleep, while children described their own sleep more variably (“fairly well,” “very well”). This divergence suggests parents underestimate mild disturbances that children themselves perceive.

Inter-Rater Agreement

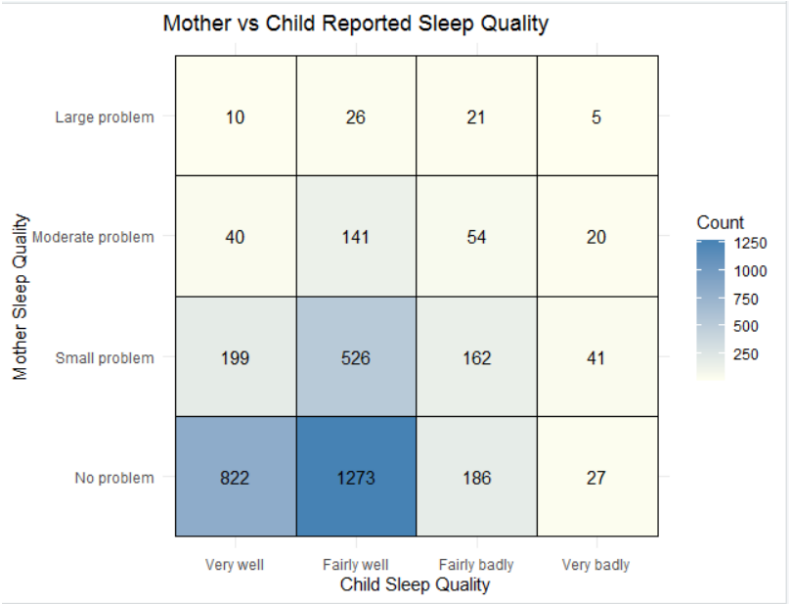


Figure 14 Cross-tabulation Heatmap of Mother vs Child Reported Sleep Quality

```
Cohen's Kappa for 2 Raters (Weights: squared)

Subjects = 3553
  Raters = 2
   Kappa = -0.00252

       z = -1.49
 p-value = 0.137
```

Table 16 Cohen's Kappa Test for Inter-Rater Agreement

Cross-tabulation and Cohen's Kappa (κ = –0.0025, p = .137) revealed no significant agreement between mother- and child-reported sleep quality. Mothers tended to under-report severity when children self-identified poor sleep. This highlights that parental assessments may not capture subjective sleep experiences and supports the need for self-report inclusion in adolescent sleep research.

Predictive Modelling of Sleep Quality

Mother-Reported Sleep Quality

```
Call:
polr(formula = MotherSleepQ ~ relevel(ethnicity, ref = "Sole European") +
    NZDEP + Rurality + TransNB + DEPRESS_SCORE_10_Y12C + PAS_T_SCORE2_Y12C +
    resilience_score, data = data_y12_nz, Hess = TRUE)
```

	OR	Lower_95_CI	Upper_95_CI
relevel(ethnicity, ref = "Sole European")Maori	0.956	0.805	1.134
relevel(ethnicity, ref = "Sole European")Pasifika	0.722	0.562	0.923
NZDEP2	0.901	0.737	1.100
NZDEP3	0.854	0.689	1.057
NZDEP4	0.858	0.681	1.080
NZDEP5	1.009	0.793	1.283
RuralityRural	1.015	0.850	1.210
TransNBGirl	1.084	0.930	1.265
TransNBTrans/NB	1.032	0.835	1.273
DEPRESS_SCORE_10_Y12C	1.067	1.048	1.086
PAS_T_SCORE2_Y12C	1.002	0.993	1.011
resilience_score	0.949	0.908	0.992

Table 17 Odds Ratios (OR) and 95% Confidence Intervals for Predictors of Mother-Reported Sleep Quality (Age 12)

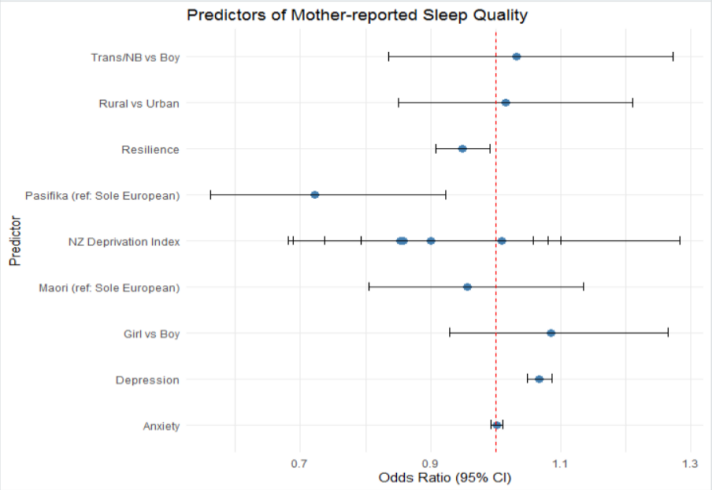


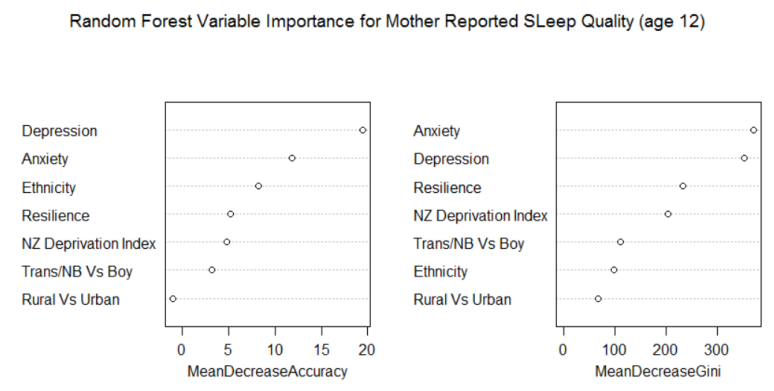
Figure 15 Predictors of Mother-Reported Sleep Quality

Ordinal logistic regression identified three significant predictors: Pasifika ethnicity, depression, and resilience. Pasifika children were 28 % less likely to be rated as having poor sleep (OR = 0.72, 95 % CI [0.56, 0.92]). Each one-point increase in depression raised odds of poor sleep by 7 % (OR = 1.07), while each resilience points reduced odds by 5 % (OR = 0.95). Psychological wellbeing variables thus dominate over socioeconomic ones. Model diagnostics confirmed proportional-odds validity (minor NZDEP deviation deemed negligible).

Random-Forest Analysis (Mother Report)

```
Call:
 randomForest(formula = MotherSleepQ ~ ethnicity + NZDEP + Rurality +      TransNB + DEPRESS_SCORE_10_Y12
C + PAS_T_SCORE2_Y12C + resilience_score,      data = data_rf, ntree = 500, mtry = 3, importance = TRUE)
               Type of random forest: classification
                     Number of trees: 500
No. of variables tried at each split: 3

        OOB estimate of  error rate: 38.46%
```



```
                     MeanDecreaseAccuracy
ethnicity                        8.225437
NZDEP                            4.782401
Rurality                        -1.068833
TransNB                          3.135060
DEPRESS_SCORE_10_Y12C           19.428914
PAS_T_SCORE2_Y12C               11.824241
resilience_score                 5.211112
                     MeanDecreaseGini
ethnicity                    98.42066
NZDEP                       202.63700
Rurality                     66.56143
TransNB                     110.81177
DEPRESS_SCORE_10_Y12C       352.68682
PAS_T_SCORE2_Y12C           370.57100
resilience_score            231.83418
```

Figure 16 Random Forest Model and Variable Importance for Predicting Mother-Reported Sleep Quality (Age 12)

The random-forest classifier (OOB error = 38.5 %) reinforced these findings. Depression and anxiety showed the greatest mean decreases in accuracy (19.4 and 11.8), followed by resilience. Ethnicity and NZDEP had smaller contributions. The model demonstrates that emotional distress and coping capacity, rather than demographic variables, best predict mothers' perceptions of sleep quality.

Child-Reported Sleep Quality

```
Call:
polr(formula = ChildsleepQ ~ relevel(ethnicity, ref = "Sole European") +
    NZDEP + Rurality + TransNB + DEPRESS_SCORE_10_Y12C + PAS_T_SCORE2_Y12C +
    resilience_score, data = data_y12_nz, Hess = TRUE)
```

	OR	Lower_95_CI	Upper_95_CI
relevel(ethnicity, ref = "Sole European")Maori	0.746	0.634	0.879
relevel(ethnicity, ref = "Sole European")Pasifika	0.485	0.387	0.608
NZDEP2	0.959	0.792	1.160
NZDEP3	0.966	0.790	1.181
NZDEP4	0.867	0.700	1.075
NZDEP5	0.683	0.543	0.859
RuralityRural	0.994	0.841	1.174
TransNBGirl	0.966	0.836	1.117
TransNBTrans/NB	1.215	0.992	1.488
DEPRESS_SCORE_10_Y12C	1.254	1.230	1.279
PAS_T_SCORE2_Y12C	0.989	0.981	0.997
resilience_score	0.954	0.915	0.995

Table 18 Odds Ratios (OR) and 95% Confidence Intervals for Predictors of Child-Reported Sleep Quality (Age 12)

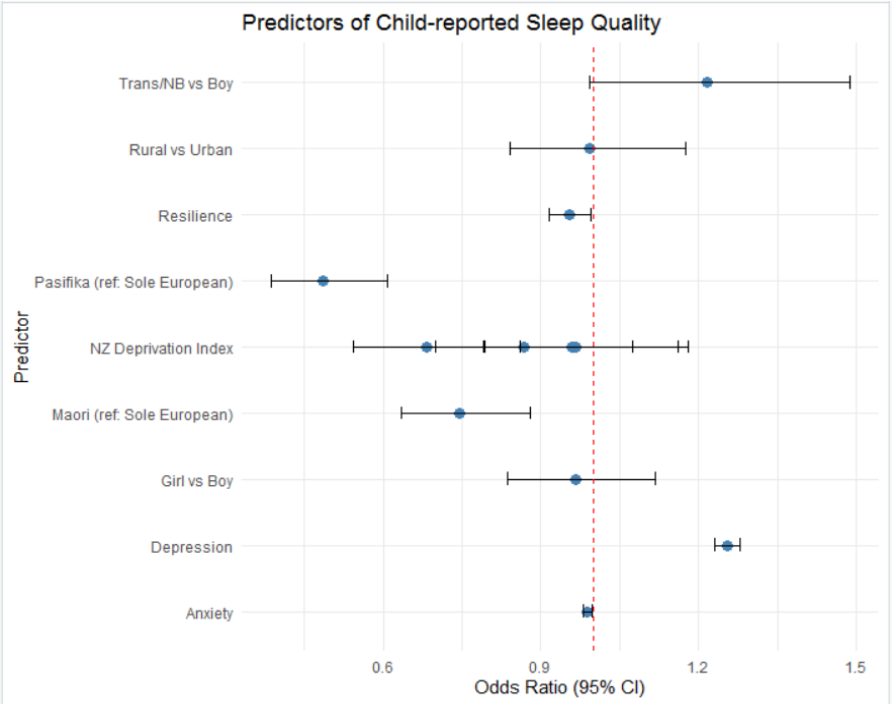


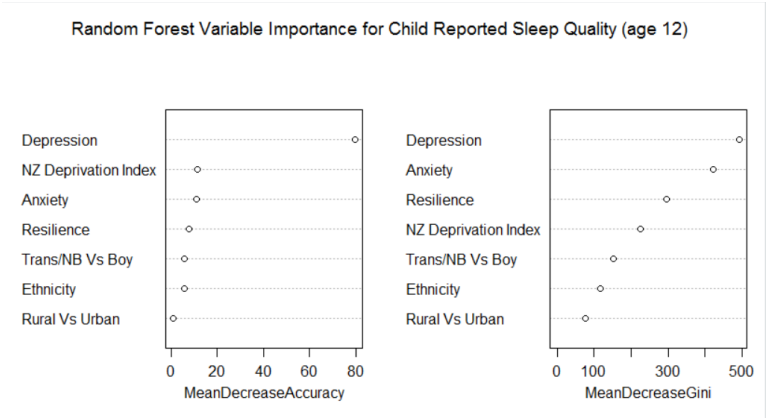
Figure 17 Predictors of Child-Reported Sleep Quality

Child self-rates told a similar but stronger story. Ordinal logistic regression indicated Māori children had 25 % lower odds of poor sleep (OR = 0.75), and Pasifika 52 % lower (OR = 0.49) relative to Sole Europeans. Depression substantially increased odds of poor sleep (OR = 1.25), while resilience again offered protection (OR = 0.95). Anxiety exerted a small but significant negative influence (OR = 0.99). Socioeconomic, rurality, and gender variables were nonsignificant, confirming that psychological factors dominate.

Random-Forest Analysis (Child Report)

```
Call:
 randomForest(formula = ChildsleepQ ~ ethnicity + NZDEP + Rurality +      TransNB +
DEPRESS_SCORE_10_Y12C + PAS_T_SCORE2_Y12C + resilience_score,      data = data_rf2,
ntree = 500, mtry = 3, importance = TRUE)
               Type of random forest: classification
                     Number of trees: 500
No. of variables tried at each split: 3

        OOB estimate of  error rate: 45.11%
```




```
                     MeanDecreaseAccuracy
ethnicity                       5.8274132
NZDEP                          11.5227648
Rurality                        0.7686475
TransNB                         5.9433113
DEPRESS_SCORE_10_Y12C          79.7711810
PAS_T_SCORE2_Y12C              11.0659244
resilience_score                7.8912453
                     MeanDecreaseGini
ethnicity                   117.56862
NZDEP                       225.37721
Rurality                     76.14719
TransNB                     153.02337
DEPRESS_SCORE_10_Y12C       493.98437
PAS_T_SCORE2_Y12C           421.83478
resilience_score            295.92677
```

Figure 18 Random Forest Model and Variable Importance for Predicting Child-Reported Sleep Quality (Age 12)

Machine-learning analysis further validated these trends. Depression emerged as the top predictor, with highest mean decrease in accuracy (79.77) and node purity (493.98). Anxiety and resilience followed. NZ Deprivation Index had moderate influence, while ethnicity, rurality, and gender were minor. Thus, self-perceived sleep quality in early adolescence is shaped primarily by emotional wellbeing rather than structural demographics.

Integrated Summary of Findings

1. Developmental Decline in Sleep: From 2 to 12 years, children's total daily sleep decreased by roughly 3 hours, mirroring typical maturation processes and social-environmental demands.

2. Ethnic Disparities: Māori and Pasifika children consistently slept less than Sole Europeans, though their longitudinal decline was shallower, narrowing differences by age 12.

3. Household Determinants: Extended family structures, food insecurity, and private rental tenure significantly reduced sleep duration, whereas income and rurality were largely irrelevant.
4. Sleep Problems: Prevalence declined sharply after early childhood, suggesting increasing sleep stability and regulation.
5. Psychological Correlates: At age 12, shorter sleep was linked with higher depression and anxiety and lower resilience—confirming bidirectional interplay between sleep and emotional wellbeing.
6. Parent–Child Perception Gap: Mothers generally underestimated children's sleep difficulties; inter-rater agreement was statistically absent.
7. Predictive Modelling: Across both regression and machine-learning approaches, psychological variables (especially depression) emerged as dominant predictors of perceived sleep quality, surpassing demographic or socioeconomic factors.
8. Temporal Relationships: Depression at 12 was not associated with earlier sleep duration, implying that mental-health differences arise from later developmental contexts rather than early sleep trajectories.

Overall Interpretation

Collectively, the findings highlight the multifactorial nature of children's sleep patterns. Biological maturation drives the primary age-related decline, yet social and cultural contexts modulate its extent. Ethnic and family-structural effects demonstrate that sleep health cannot be disentangled from living environments and cultural practices. By adolescence, psychological wellbeing becomes the principal correlation of both objective and subjective sleep outcomes, underscoring a developmental transition from environmental to emotional determinants.

These outcomes contribute to the national understanding of sleep health disparities in Aotearoa New Zealand, emphasizing that interventions should combine culturally sensitive family-level supports in early childhood with mental-health-focused strategies in adolescence.

Discussion and Critical Analysis

Interpretation of Key Findings

The decline in sleep duration with age aligns with biological maturation, but ethnic disparities reflect structural inequities like overcrowding (Howden-Chapman et al., 2021). Shorter sleep predicting poorer mental health supports global evidence (Carpenter et al., 2021) though the lack of early sleep-depression links suggests later

social stressors (e.g., school demands) are critical. Cultural practices like co-sleeping may buffer or exacerbate sleep, warranting further exploration (Paine et al., 2022).

Strengths and Limitations

Strengths include the longitudinal design, large sample, and cultural grounding, enhancing generalizability. Limitations encompass self-reported data, lacking objective measures (e.g., actigraphy), and potential recall bias. The modest R^2 values (e.g., 0.018 for sleep duration models) indicate unmeasured environmental factors.

Implications for GUiNZ and Policy

Findings underscore the need for whānau-centered interventions, aligning with GUiNZ's equity focus. Enhanced data collection on sleep environments and longitudinal mental health tracking could refine future analyses. Policy implications suggest targeting sleep health in deprived areas to reduce mental health disparities.

Conclusions and Recommendations

Summary of Findings

This study examined longitudinal sleep patterns and their psychosocial correlates among children participating in the Growing Up in New Zealand (GUiNZ) cohort. Across the four developmental stages (ages 2, 4, 8, and 12 years), total sleep duration declined markedly from an average of 12.3 hours at age 2 to 9.2 hours at age 12, reflecting the natural maturation of sleep–wake regulation and increasing environmental demands such as schooling and social engagement.

Ethnic differences were consistently observed. Māori and Pasifika children recorded shorter average sleep durations compared with Sole European peers, a disparity partly explained by household factors including family structure, food insecurity, and housing tenure. Children living in extended or non-kin households or in private rentals tended to sleep less, suggesting that family complexity and residential instability may constrain optimal sleep environments.

Sleep duration was closely associated with psychological wellbeing at age 12. Shorter sleep was linked to higher levels of depression and anxiety, whereas longer sleep correlated with greater resilience. However, analyses of earlier sleep patterns (ages 2–8) found no predictive association with later depression, indicating that emotional wellbeing during pre-adolescence is shaped more by contemporaneous social and psychological factors than by early-life sleep behaviour. These findings underscore the importance of contextual and cultural determinants of sleep, as well as their implications for children's mental health and equity outcomes.

Recommendations for Intervention and Policy

1. Culturally Tailored Sleep Programs:
   Develop sleep health resources that reflect Māori and Pasifika cultural norms, including co-sleeping practices, bedtime routines, and collective caregiving structures. Co-designing such interventions with whānau and community leaders will enhance cultural resonance and uptake.

2. Policy Integration:
   Embed sleep health within national mental-health and child-wellbeing strategies, particularly in areas with high socioeconomic deprivation or large Māori and Pasifika populations. Integrating sleep promotion into primary healthcare, schools, and community services can help address disparities early.

3. Research–Practice Collaboration:
   Strengthen partnerships between academic institutions, iwi authorities, and Pacific organisations to design, implement, and evaluate culturally grounded interventions. Shared governance and data sovereignty principles should guide all initiatives.

4. Further Research Expansion:
   Future iterations of GUiNZ or related studies should incorporate objective sleep measures (e.g., actigraphy or wearable technology) to complement parental and self-reports, improving accuracy and comparability across contexts. Collecting information on cultural moderators—such as whānau support, collective living, and identity—will deepen understanding of how cultural frameworks influence sleep and wellbeing.

Future Research Directions

Future investigations should move beyond duration to examine sleep quality and variability using both subjective and physiological indicators. Extending longitudinal follow-ups into adolescence and early adulthood will allow assessment of how childhood sleep patterns relate to long-term emotional, cognitive, and educational outcomes. Additionally, exploring cultural mediators such as whānau cohesion, community belonging, and bilingual or bicultural identity could clarify protective factors that buffer against poor sleep or mental distress. By integrating cultural, environmental, and biological perspectives, future research can inform more equitable health policies and contribute to holistic models of child wellbeing in Aotearoa New Zealand.

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Appendix

Disclaimer

Auckland University of Technology
Master of Analytics
Research Project

Disclaimer:

Clients should note the general basis upon which the Auckland University of Technology undertakes its student projects on behalf of external sponsors:

While all due care and diligence will be expected to be taken by the students, (acting in data analytics, statistics, research or other professional capacities), and the Auckland University of Technology, and student efforts will be supervised by experienced AUT lecturers, it must be recognised that these projects are undertaken in the course of student instruction. There is therefore no guarantee that students will succeed in their efforts.

This inherently means that the client assumes a degree of risk. This is part of an arrangement, which is intended to be of mutual benefit. On completion of the project it is hoped that the client will receive a professionally documented project report, while the students are exposed to live external environments and problems, in a realistic project and customer context.

In consequence of the above, the students, acting in their assigned professional capacities and the Auckland University of Technology, disclaim responsibility and offer no warranty in respect of outcomes of the project both in relation to their use and results from their use.